Project: Worked examples
Title: Member checks (EN 1993-1-1)
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1. Scope and method

Resistance checks of a steel member of class 1, 2 or 3 cross-section: tension (6.2.3), compression (6.2.4) and flexural buckling about y and z (6.3.1), bending about y and z (6.2.5), shear with the high-shear reduction (6.2.6, 6.2.8), the linear cross-section interaction (6.2.1(7)) and a simplified member interaction (6.3.3 with user-entered k factors). Lateral-torsional buckling and class 4 sections are outside this sheet.

WHAT IT CALCULATES	RELATION
Axial plastic resistance	$N_{pl,Rd} = A\, f_y/\gamma_{M0}$
Bending resistance, plastic modulus for class 1–2 and elastic for class 3	$M_{c,Rd} = W\, f_y/\gamma_{M0}$
Shear resistance	$V_{pl,Rd} = A_v\, f_y/(\sqrt{3}\,\gamma_{M0})$
Bending resistance reduced by high shear, when $V_{Ed} > 0.5\, V_{pl,Rd}$	$\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$. Class 1–2 I-section with h, t_w, t_f entered: $M_{y,V,Rd} = (W_{pl,y} - \rho\, A_w^2/(4\, t_w))\, f_y/\gamma_{M0}$, $A_w = (h - 2t_f)\, t_w$, and $M_{z,c,Rd}$ is not reduced; otherwise the whole resistance × $(1 - \rho)$
Elastic critical load and non-dimensional slenderness	$N_{cr} = \pi^2 EI/L_{cr}^2$; $\bar{\lambda} = \sqrt{(A\, f_y/N_{cr})}$
Buckling reduction factor and buckling resistance	$\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$; $N_{b,Rd} = \chi\, A\, f_y/\gamma_{M1}$
Cross-section check, linear interaction	$N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$
Member check, axial force and bending with buckling (6.61, 6.62)	$N_{Ed}/N_{b,Rd,min} + k_{yy}\, M_{y,Ed}/(W_y\, f_y/\gamma_{M1}) + k_{zz}\, M_{z,Ed}/(W_z\, f_y/\gamma_{M1}) \le 1$, the characteristic bending resistance M_{Rk} over γ_{M1} without the shear reduction

Basis: Tension[1](6.2.3); Compression[1](6.2.4); Bending moment[1](6.2.5); Shear[1](6.2.6); Bending and shear[1](6.2.8); Linear summation of utilisations[1](6.2.1(7)); Uniform members in compression: buckling resistance, buckling curves[1](6.3.1.1, 6.3.1.2); Uniform members in bending and axial compression[1](6.3.3); Imperfection factors for the buckling curves[2](§6.3.1).

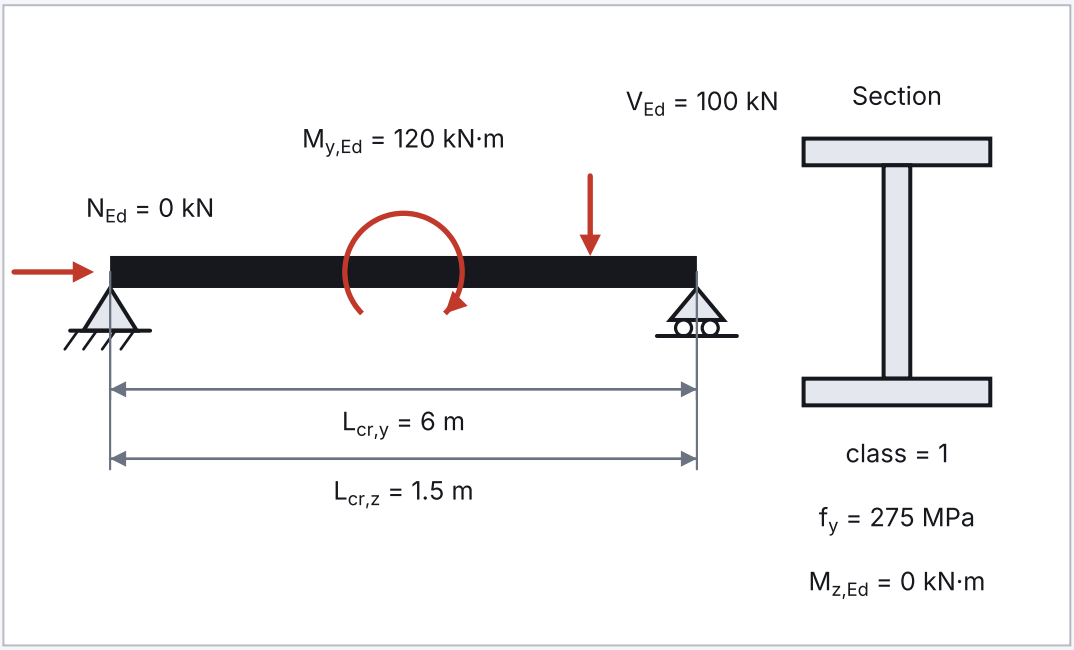


Figure 1 — A steel member under axial force, bending and shear, its buckling lengths about both axes, and its section. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_{Ed}	Design axial force, TENSION positive and compression negative (the opposite sign to S6's N_{Ed}, which is compression positive; S2's N_{e1} is tension positive)	0	kN	reference: S2 frame sheet, two-span beam, ULS	-100000 … 100000 kN
$M_{y,Ed}$	Design moment about y-y (major)	120	kN·m	reference: S2 frame sheet, two-span beam, ULS (rounded)	0 … 100000 kN·m
$M_{z,Ed}$	Design moment about z-z (minor)	0	kN·m	reference: S2 frame sheet	0 … 100000 kN·m
V_{Ed}	Design shear force	100	kN	reference: S2 frame sheet, two-span beam, ULS (rounded)	0 … 100000 kN
f_y	Yield strength (for the thickness)	275	MPa	spec: BS EN 10025-2 S275, t ≤ 16 mm	200 … 700 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	190 … 220 GPa
γ_{M0}	Partial factor γ_{M0} (cross-section resistance)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
γ_{M1}	Partial factor γ_{M1} (member buckling)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$section_{class}$	Cross-section class	1		assumption: UB 305×165×40 S275 is class 1 in bending (Blue Book).	any
A	Area A	51.3	cm^2	reference: S1 section sheet / Blue Book UB 305×165×40	1 … 100000 cm^2
$W_{pl,y}$	$W_{pl,y}$	623	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,y}$	$W_{el,y}$	560	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{pl,z}$	$W_{pl,z}$	142	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,z}$	$W_{el,z}$	92.6	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
I_y	I_y	8503	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
I_z	I_z	764	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
h	Depth h (shear-area helper; leave blank when A_v is given)	303.4	mm	spec: BS EN 10365 UB 305×165×40	1 … 5000 mm
b	Width b (helper)	165	mm	spec: BS EN 10365 UB 305×165×40	1 … 5000 mm
t_w	Web thickness (helper)	6	mm	spec: BS EN 10365 UB 305×165×40	0.1 … 500 mm
t_f	Flange thickness (helper)	10.2	mm	spec: BS EN 10365 UB 305×165×40	0.1 … 500 mm
r	Root radius (helper)	8.9	mm	spec: BS EN 10365 UB 305×165×40	0 … 500 mm
η	Shear-area factor η	1		spec: UK NA to BS EN 1993-1-5 NA.2.4	1 … 1.2
$L_{cr,y}$	Buckling length about y-y	6	m	assumption: Span 6 m.	0.1 … 100 m
$L_{cr,z}$	Buckling length about z-z	1.5	m	assumption: Compression flange restrained at 1.5 m by the deck.	0.1 … 100 m
$curve_y$	Buckling curve about y-y	a		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b > 1.2, t_f ≤ 40, y-y	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$curve_z$	Buckling curve about z-z	b		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b > 1.2, t_f ≤ 40, z-z	any
k_{yy}	Interaction factor k_{yy} (member check)	1		assumption: No axial force; interaction factors irrelevant.	0.5 … 2
k_{zz}	Interaction factor k_{zz} (member check)	1		assumption: No axial force; interaction factors irrelevant.	0.5 … 2
ltb	Compression flange against lateral-torsional buckling (not restrained: outside this sheet)	Restrained		default	any
γ_{M2}	Partial factor γ_{M2} (net-section fracture)	1.1		default	1 … 1.35

3. Assumptions

A1. The cross-section class is as stated by the user (1, 2 or 3); class 4 effective properties are not derived here. (6.2.2.5 effective widths are outside this sheet.)
A2. The compression flange is restrained against lateral-torsional buckling, as the input ltb says. (6.3.2 is outside this sheet: a member that is not restrained is checked on the LTB interaction sheet.)
A3. A k left at 1.0 under $N_{Ed}/N_{b,Rd}$ above 0.3 is replaced by the Annex A/B upper bound $C_m(1 + 0.8n)$. (6.3.3 simplified: the smaller flexural-buckling reduction is used on the axial term and the factors are not derived.)
A4. Imperfection factors per curve are the textbook values and may be overridden; partial factors default to 1.0 (UK National Annex) and may be overridden. (User-editable with provenance.)
A5. $section_{class}$ (Cross-section class) is taken as 1. (UB 305×165×40 S275 is class 1 in bending (Blue Book).)
A6. $L_{cr,y}$ (Buckling length about y-y) is taken as 6 m. (Span 6 m.)
A7. $L_{cr,z}$ (Buckling length about z-z) is taken as 1.5 m. (Compression flange restrained at 1.5 m by the deck.)
A8. k_{yy} (Interaction factor k_{yy} (member check)) is taken as 1. (No axial force; interaction factors irrelevant.)
A9. k_{zz} (Interaction factor k_{zz} (member check)) is taken as 1. (No axial force; interaction factors irrelevant.)
A10. The compression flange is restrained against lateral-torsional buckling, as the input ltb says. (Input ltb.)

4. Working

[28] $r_h = r$
$= 8.9\ mm$
$= 8.9\ mm$
Root radius used by the helper

[29] $A_{v,rolled} = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f$ [1](6.2.6(3))
$= 51.3\ cm^2 - 2 \times 165\ mm \times 10.2\ mm + (6\ mm + 2 \times 8.9\ mm) \times 10.2\ mm$
$= 20.068\ cm^2$
Rolled I shear area, load parallel to the web

[30] $A_{v,min} = \eta * (h - 2 * t_f) * t_w$ [1](6.2.6(3))
$= 1 \times (303.4\ mm - 2 \times 10.2\ mm) \times 6\ mm$
$= 16.98\ cm^2$

Lower bound η h_w t_w

[31] $A_v = \max(A_{v,rolled}, A_{v,min})$
$= \max(20.068\ cm^2, 16.98\ cm^2)$
$= 20.068\ cm^2$

Shear area

[32] $N_{pl,Rd} = A * f_y / \gamma_{M0}$ [1](6.2.3(2), 6.2.4(2))
$= 51.3\ cm^2 \times 275\ MPa / 1$
$= 1410.8\ kN$

Plastic axial resistance

[33] $W_y = W_{pl,y}$ [1](6.2.5(2))
$= 623\ cm^3$
$= 623\ cm^3$

Section modulus about y (plastic, class 1–2)

[34] $W_z = W_{pl,z}$ [1](6.2.5(2))
$= 142\ cm^3$
$= 142\ cm^3$

Section modulus about z (plastic)

[35] $V_{pl,Rd} = A_v * f_y / (\sqrt{3} * \gamma_{M0})$ [1](6.2.6(2))
$= 20.068\ cm^2 \times 275\ MPa / (\sqrt{3} \times 1)$
$= 318.62\ kN$

Plastic shear resistance

[36] $u_V = V_{Ed} / V_{pl,Rd}$
$= 100\ kN / 318.62\ kN$
$= 0.31386$

Shear utilisation

[37] $\rho = 0$ [1](6.2.8(2))
$= 0$

Shear reduction factor ($V_{Ed} \le 0.5\ V_{pl,Rd}$)

[38] $M_{y,c,Rd} = (1 - \rho) * W_y * f_y / \gamma_{M0}$ [1](6.2.5(2), 6.2.8)
$= (1 - 0) \times 623\ cm^3 \times 275\ MPa / 1$
$= 171.32\ kN{\cdot}m$

Bending resistance about y

[39] $M_{z,c,Rd} = (1 - \rho) * W_z * f_y / \gamma_{M0}$ [1](6.2.5(2))
$= (1 - 0) \times 142\ cm^3 \times 275\ MPa / 1$
$= 39.05\ kN{\cdot}m$

Bending resistance about z

[40] $u_{M,y} = M_{y,Ed} / M_{y,c,Rd}$
$= 120\ kN{\cdot}m / 171.32\ kN{\cdot}m$
$= 0.70042$

Bending utilisation, y

[41] $u_{M,z} = M_{z,Ed} / M_{z,c,Rd}$
$= 0\ kN{\cdot}m / 39.05\ kN{\cdot}m$
$= 0$

Bending utilisation, z

[42] $u_N = \mathrm{abs}(N_{Ed}) / N_{pl,Rd}$
$= \mathrm{abs}(0\ kN) / 1410.8\ kN$
$= 0$

Tension utilisation

[43] $u_{section} = u_N + u_{M,y} + u_{M,z}$ [1](6.2.1(7))
$= 0 + 0.70042 + 0$
$= 0.70042$

Linear cross-section interaction

[46] α_y = imperfection factor, curve a [1](6.3.1.2)
= Designers' Guide §6.3.1
$= 0.21$

Imperfection factor about y

[47] α_z = imperfection factor, curve b [1](6.3.1.2)

= Designers' Guide §6.3.1

= 0.34

Imperfection factor about z

[48] $N_{cr,y}$ = pi^2 * E * I_y / $L_{cr,y}$^2 [1](6.3.1.2(1))

= π^2 × 210 GPa × 8503 cm⁴ / (6 m)^2

= 4895.4 kN

Elastic critical force about y

[49] λ_y = sqrt(A * f_y / $N_{cr,y}$) [1](6.3.1.2(1))

= sqrt(51.3 cm² × 275 MPa / 4895.4 kN)

= 0.53682

Non-dimensional slenderness about y

[50] Φ_y = 0.5 * (1 + α_y * (λ_y - 0.2) + λ_y^2) [1](6.3.1.2(1))

= 0.5 × (1 + 0.21 × (0.53682 - 0.2) + 0.53682^2)

= 0.67946

Φ about y

[51] χ_y = min(1, 1 / (Φ_y + sqrt(Φ_y^2 - λ_y^2))) [1](6.3.1.2(1))

= min(1, 1 / (0.67946 + sqrt(0.67946^2 - 0.53682^2)))

= 0.91244

Reduction factor χ about y

[52] $N_{b,y,Rd}$ = χ_y * A * f_y / γ_{M1} [1](6.3.1.1(3))

= 0.91244 × 51.3 cm² × 275 MPa / 1

= 1287.2 kN

Buckling resistance about y

[53] $N_{cr,z}$ = pi^2 * E * I_z / $L_{cr,z}$^2 [1](6.3.1.2(1))

= π^2 × 210 GPa × 764 cm⁴ / (1.5 m)^2

= 7037.7 kN

Elastic critical force about z

[54] λ_z = sqrt(A * f_y / $N_{cr,z}$) [1](6.3.1.2(1))

= sqrt(51.3 cm² × 275 MPa / 7037.7 kN)

= 0.44772

Non-dimensional slenderness about z

[55] Φ_z = 0.5 * (1 + α_z * (λ_z - 0.2) + λ_z^2) [1](6.3.1.2(1))

= 0.5 × (1 + 0.34 × (0.44772 - 0.2) + 0.44772^2)

= 0.64234

Φ about z

[56] χ_z = min(1, 1 / (Φ_z + sqrt(Φ_z^2 - λ_z^2))) [1](6.3.1.2(1))

= min(1, 1 / (0.64234 + sqrt(0.64234^2 - 0.44772^2)))

= 0.90667

Reduction factor χ about z

[57] $N_{b,z,Rd}$ = χ_z * A * f_y / γ_{M1} [1](6.3.1.1(3))

= 0.90667 × 51.3 cm² × 275 MPa / 1

= 1279.1 kN

Buckling resistance about z

[58] $N_{b,Rd}$ = min($N_{b,y,Rd}$, $N_{b,z,Rd}$)

= min(1287.2 kN, 1279.1 kN)

= 1279.1 kN

Governing buckling resistance

[59] $u_{buckling}$ = 0 [1](6.3.1.1(1))

= 0

Flexural buckling utilisation (tension: not applicable)

[60] $M_{y,b,Rd}$ = W_y * f_y / γ_{M1} [1](6.3.3, Eq. 6.61)

= 623 cm³ × 275 MPa / 1

= 171.32 kN·m

Characteristic bending resistance about y over γ_{M1} (member check, no shear reduction)

[61] $M_{z,b,Rd} = W_z * f_y / \gamma_{M1}$ [1](6.3.3, Eq. 6.62)

$= 142\ cm^3 \times 275\ MPa / 1$

$= 39.05\ kN \cdot m$

Characteristic bending resistance about z over γ_{M1} (member check, no shear reduction)

[62] $u_{member} = u_{M,y} + u_{M,z}$ [1](6.3.3(4))

$= 0.70042 + 0$

$= 0.70042$

Simplified member interaction

5. Results

SYMBOL	VALUE	DESCRIPTION
A_v	20.07 cm²	Shear area
$N_{pl,Rd}$	1410.8 kN	$N_{pl,Rd}$
$V_{pl,Rd}$	318.62 kN	$V_{pl,Rd}$
ρ	0	ρ
$M_{y,c,Rd}$	171.32 kN·m	$M_{y,c,Rd}$
$M_{z,c,Rd}$	39.05 kN·m	$M_{z,c,Rd}$
$M_{y,b,Rd}$	171.32 kN·m	$M_{y,Rk}/\gamma_{M1}$
$M_{z,b,Rd}$	39.05 kN·m	$M_{z,Rk}/\gamma_{M1}$
$N_{cr,y}$	4895.4 kN	$N_{cr,y}$
$N_{cr,z}$	7037.7 kN	$N_{cr,z}$
λ_y	0.53682	$\bar{\lambda}_y$
λ_z	0.44772	$\bar{\lambda}_z$
χ_y	0.91244	χ_y
χ_z	0.90667	χ_z
$N_{b,y,Rd}$	1287.2 kN	$N_{b,y,Rd}$
$N_{b,z,Rd}$	1279.1 kN	$N_{b,z,Rd}$
$N_{b,Rd}$	1279.1 kN	$N_{b,Rd}$
u_N	0.00	Axial utilisation
$u_{M,y}$	0.70	Bending y utilisation (governing)
$u_{M,z}$	0.00	Bending z utilisation
u_V	0.31	Shear utilisation
$u_{section}$	0.70	Cross-section interaction
$u_{buckling}$	0.00	Buckling utilisation
u_{member}	0.70	Member interaction

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Bending y $M_{y,Ed} \le M_{y,c,Rd}$	$u_{M,y} = 0.70$	$u_{M,y} \le 1$	0.70	[1](6.2.5)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Cross-section interaction	$u_{section} = 0.70$	$u_{section} \le 1$	0.70	[1](6.2.1(7))
PASS	Shear $V_{Ed} \le V_{pl,Rd}$	$u_V = 0.31$	$u_V \le 1$	0.31	[1](6.2.6)
PASS	Tension $N_{Ed} \le N_{pl,Rd}$	$u_N = 0.00$	$u_N \le 1$	0.00	[1](6.2.3)
PASS	Bending z $M_{z,Ed} \le M_{z,c,Rd}$	$u_{M,z} = 0.00$	$u_{M,z} \le 1$	0.00	[1](6.2.5)

7. Applicability

None.

Defaults used (inputs left empty):
ltb: not entered, so the default of 'Restrained' is used.
γ_{M2}: not entered, so the default of 1.1 is used.

8. Verification

`Template struct.en1993-member@1.4.0 Content hash fdf55f0878a74ece Engine calc-engine/1.0.0`
`Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0`

9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. Gardner L, Nethercot DA. Designers' Guide to EN 1993-1-1: Eurocode 3: Design of Steel Structures, General Rules and Rules for Buildings. Thomas Telford; 2005.